

An All-Out Test to Determine Finger Flexor Critical Force in Rock Climbers

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Purpose: The fatigue resistance of the finger flexors is known to be a key determinant of climbing performance. This study set out to establish the association between the single all-out assessment of finger flexor critical force (ff-CF) and the impulse above CF (W') on climbing performance (self-reported sport and boulder climbing ability). **Methods:** A total of 129 subjects completed an assessment of dominant arm ff-CF, comprised of a series of rhythmic isometric maximum voluntary contractions (CF defined as mean end-test force [in kilograms]; W' impulse above CF [in kilogram second]). **Results:** The ff-CF protocol resulted in the same force decay to a plateau seen in previous isometric critical torque and critical force tests. Linear regression analysis, adjusting for sex, revealed that CF percentage of body mass explained 61% of sport and 26% of bouldering performance and W' per kilogram body mass explained 7% sport and 34% bouldering performance. A combined model of CF as a percentage of body mass and W' per kilogram body mass, after adjustment for sex differences, was able to explain 66% of sport climbing and 44% of bouldering performance. **Conclusions:** The results illustrate the relevance of the CF threshold in describing the fatigue resistance of the finger flexors of rock climbers. Given ff-CF ability to describe a considerable proportion of variance in sport climbing and bouldering ability, the authors expect it to become a common test used by coaches for understanding exercise tolerance and for determining optimal training prescription.

Keywords: aerobic capacity, exercise testing, fatigue, criterion validity, endurance testing

Rock climbing requires repeated isometric contractions of the finger flexors, which are responsible for the flexion of the metacarpophalangeal and interphalangeal joints.¹ These contractions cause regular periods of ischemia in the forearm muscles; the extent of this ischemia and the subsequent recovery from it has been shown to differentiate between ability groups of rock climbers,² as well as disciplines,³ and is likely to be a trainable characteristic.⁴ As such, the fatigue resistance of the finger flexors is considered one of the most important factors in climbing performance, and it is common for climbers to use a “fingerboard” for the training of the finger flexors.

While methods for the determination of maximal finger flexor strength in rock climbers have been described in the literature,⁵ no tests to determine functional aerobic metabolic capacity, delineating steady and nonsteady states, existed until recently. Our research group recently presented the first data on the sensitivity of a 3 session force-time model for determining critical force (CF) in the finger flexors of rock climbers,⁶ along with the work capacity that may be completed above CF, termed W' (often described as the “energy store” component⁷). CF is the maximal force output that still results in a metabolic steady state characterized by a plateau in $\dot{V}O_2$ and in inorganic phosphates.^{7,8} Our study demonstrated the sensitivity of a simple test for the determination of CF, using equipment readily available to climbers and coaches in most climbing gyms. However, the “traditional” method that the test

was based on, requires subjects to exercise to exhaustion multiple times under different constant workloads, often on separate days, and as a result may be less repeatable and may not always be convenient or practical.

A potential alternative approach would be to identify CF using an “all-out” test. According to the concept of critical power (CP; the isotonic equivalent of CF), if W' is fully utilized (reduced to zero), the maximum power output possible would be CP.^{9–12} In other words, if there was a method to completely deplete W' , the remaining power output should equal to CP. For example, Coats et al⁹ noted that CP was the greatest power output that could be maintained after a fatiguing cycle exercise (ie, an exercise where W' is depleted). In the 3-minute cycling test, the authors asked subjects to perform an all-out effort to deplete W' after which the power sustained was equal to CP; this has previously been termed the end-test power.^{10–12} The same relationship between CP and end-test power has been shown to be true for isometric work and thus CF.¹³

Isometric single bout, all-out CF tests have been validated for a number of synergistic muscle groups and exercise modalities, including the forearms using handgrip dynamometry.¹³ However, given the lack of specificity of handgrip dynamometry to climbing performance,^{14,15} any test of finger flexors CF (ff-CF; force analog of CP) in climbers must use work to relief ratios and hand and body positions representative of those found in the sport. Thus, the primary aim of the present study was to use a climbing-specific maximal effort rhythmic isometric fingerboard test to estimate ff-CF, in order to assess (1) how much variance in self-reported sport climbing (longer and endurance focused) and bouldering (short and powerful) climbing performance may be explained by CF and W' , (2) what is the combined contribution of CF and W' to climbing performance while adjusting for sex and (3) is there a relationship between maximal isometric finger flexor strength (MIFS) and CF.

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Methods

Subjects

A total of 129 (61 females and 68 males) active climbers of intermediate to higher elite ability (International Rock Climbing Research Association [IRCRA]) grading scale¹⁶) volunteered to participate in the study (described in Table 1). Subjects were recruited on the basis of being familiar with climbing specific forearm training and exhaustive forearm exercise using a finger-board (training at least 3 times per month for the past 3 mo), free from injury and having no known musculoskeletal, cardiovascular or respiratory diseases, or illnesses. Written informed consent and medical health questionnaires were completed prior to participation. Institutional ethical approval from the University of Derby Human Sciences Research Ethics Committee [15-1819-DGs] was granted prior to data collection, and all protocols conformed to the principles of the Declaration of Helsinki.

Self-Reported Climbing Ability. Subjects reported their red-point (climb completed after prior practice) grade for which they have completed 3 successful ascents on 3 different routes (at the grade) within the 6 months prior to data collection for both bouldering (short powerful climbs close to the ground) and sport climbing (longer more endurance-focused climbs, using ropes). Subjects also provided data on their preferred discipline in the form of the discipline they spent the most time practicing. The validity of self-reported climbing ability has previously been established by Draper et al¹⁷ and data were collected following the guidelines set out by the IRCRA.¹⁶ Participants were allocated to groups based on their preferred discipline of sport and/or boulder. Participants were assigned to a single discipline if they (1) indicated a single specialization, (2) only reported 1 grade, or (3) spent more than 75% of their time practicing a single discipline. If the climber reported between 26% and 74% of their time spent in both disciplines, they were included in both groups, unless the difference between the 2 grades exceeded ± 3 IRCRA grades. Following this, the bouldering subgroup consisted of 76 and the sport-climbing subgroup 85 subjects, with 32 included in both data sets. The grade range was 12 to 28 IRCRA (French grading scale: f6a+ to f8c+) and 15.5 to 27 IRCRA (Vermin grading scale: V2 to V12) for the sport and boulder group, respectively.

Design

In order to determine the association between a single “all-out” test of ff-CF and self-reported ability in rock climbers, subjects completed a single assessment of their maximal isometric finger flexor strength and all-out ff-CF test along with providing demographic details of their climbing history and ability. Subjects were assessed in a rested state (having performed no heavy exercise in the 24 h preceding the test), having refrained from consuming food and caffeinated drinks for 3 hours prior. Before the assessment, each subject provided health history, informed consent, demographic data, and self-reported red-point climbing ability. Subjects were briefed on the purpose of the assessment before completing their own thorough warm-up, followed by a standardized warm-up, they were familiarized with the hand position and intermittent testing protocol. Once completed, they participated in the all-out ff-CF assessment.

Methodology

Data Collection. All MIFS and ff-CF assessments were performed on a purpose-built climbing-specific dynamometer, fitted with a 20-mm deep wooden rung with 10-mm radius. All data were recorded at a frequency of 80 Hz. Prior to each assessment, the dynamometer was calibrated and between each test was returned to 0 kilogram. Force is presented in kilograms (rather than Newtons) for ease of comparison with previous literature and for easier interpretation by coaches and climbers.

Hand and Body Positioning. All tests were performed in a one-handed half-crimp position (90° flexion at the proximal interphalangeal joint with the thumb not engaged in the grip). In accordance with Baláš et al,¹⁵ subjects were instructed to “hang” with their dominant arm extended above the head (~170° to 180° shoulder flexion), maintaining a slight bend in the elbow with shoulders engaged. Body position was controlled, asking subjects to maintain level shoulders, with their chest square to the rung and the same foot as the hand being tested in front of the other (Figure 1). For both the MIFS and ff-CF tests, subjects were instructed to develop as much force on the rung as possible by “hanging” (not “pulling”) from the edge. If the subject was capable of generating force that was close to, or greater than, their own body mass, then weight was added to a climbing sit harness around their waist so that they remained with their feet on the ground.

Warm-Up. Before beginning the assessment protocol, all subjects completed a standardized self-directed warm-up consisting of 5-minute pulse-raising activity; mobilizing (walking, jogging, or skipping); 5 minutes of climbing; and a series of 7:3 seconds work to rest ratio hangs on the testing edge in a half-crimp position at 50% and 75% of perceived maximum force.

Determination of Maximal Isometric Finger Strength. Maximal isometric finger strength was determined by performing a series of 3 single-handed 5-second maximal isometric contractions on the rung. Only the dominant arm was tested. Subjects were provided with 120 seconds rest between each attempt; 120-seconds was determined to be sufficient based on extensive testing of the protocol and has been previously documented in the literature.^{5,6,18} Force (in kilograms) and time (in seconds) data were recorded continuously throughout. Peak force data are reported as absolute units (in kilograms) and relative to body mass (in percentage).

Determination of Finger Flexor CF. The determination of CF was based on the methodology of Kellawan and Tschakovsky¹³

Table 1 Subject Characteristics

Subject characteristic	Female	Male
N	61	68
Age, y	33.9 (9.5)	30 (8.6)
Body mass, kg	59.2 (6.6)	72.3 (8.8)
Height, cm	164.5 (5.8)	176.9 (7)
BMI, kg/m ²	21.9 (4.5)	23.1 (2.1)
Experience, y	8.6 (7.4)	7.4 (7.3)
Performance, sport (range: 11–28 IRCRA)	17.4 (5.9)	18.8 (7.9)
Performance, bouldering (range: 12–28 IRCRA)	19.5 (6.5)	21.1 (5.6)

Abbreviations: BMI, body mass index; IRCRA, International Rock Climbing Research Association numeric grading scale. Note: Values are presented as mean (SD).

involving fatiguing muscle actions of the finger flexors. Finger flexor CF (ff-CF) was determined by performing a series of rhythmic isometric maximum voluntary contractions on the rung, in a half-crimp position (Figure 2) with a 7:3 seconds work to rest ratio, as previously used when assessing climbers CF.⁶

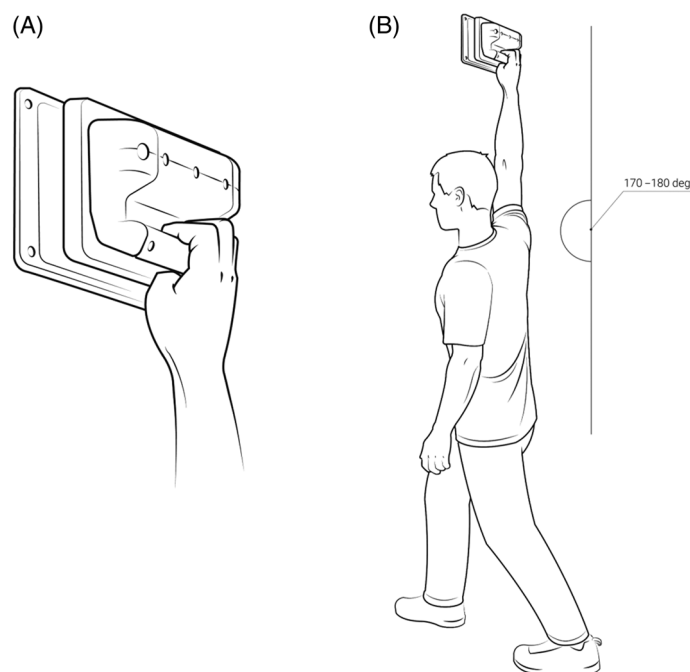


Figure 1 — (A) Half-crimp grip position on the 20-mm rung. (B) Standardized body position with level shoulders, chest square to the dynamometer, and the same foot as the hand being tested in front of the other.

During the “work” phase, subjects were instructed to produce as much force as possible while maintaining a half-crimp position. During the “rest” phase, to standardize, practice subjects were instructed to be in the anatomical position, during which they could apply climbing chalk but not shake their forearms or hands (shaking of the hands is known to aid recovery, eg, Baláš et al¹⁹). Throughout the tests, subjects were able to visually observe their force output, which was continuously displayed on a screen. Subjects were verbally encouraged to reach their maximum force on every contraction, to achieve a “square wave” during each maximal voluntary contraction (MVC), and to engage only the muscles of the forearm rather than “pulling up.” Force (in kilograms) and time (in seconds) data were recorded continuously throughout. Length (in seconds), peak and mean force (in kilograms) and the force–time impulse (in kilogram second) were determined for every contraction for all tests. CF was defined as mean end-test force, using the last 6 contractions of the test (the last 60 s¹⁰). A 1 SD cutoff was used for the inclusion of contractions in the calculation of end-test force to reduce the effects of erroneous contractions occurring due to slippage or the adjustment of hand position, which occurred in a small number of assessments. The W' was calculated as the impulse measured above the end-test force during the ff-CF test.

Initially, ff-CF assessments were 30 contractions in length (5 min; $n = 46$); however, it was observed that subjects were able to achieve a plateau in their end-test force output after 24 contractions (4 min; $n = 83$). Given the maximal nature of the testing, the discomfort caused when asking subjects to complete maximal contractions and the possibility of injury when performing maximal contractions when highly fatigued, the decision was made to reduce the testing time to 4 minutes for subsequent assessments ($n = 83$). While the data presented uses end-test force from the last 60 seconds of both 5- and 4-minute tests, cutting off the data at 4 minutes for all assessments does not produce materially different results or change the conclusions of the paper. Indeed, for the 5-minute tests,

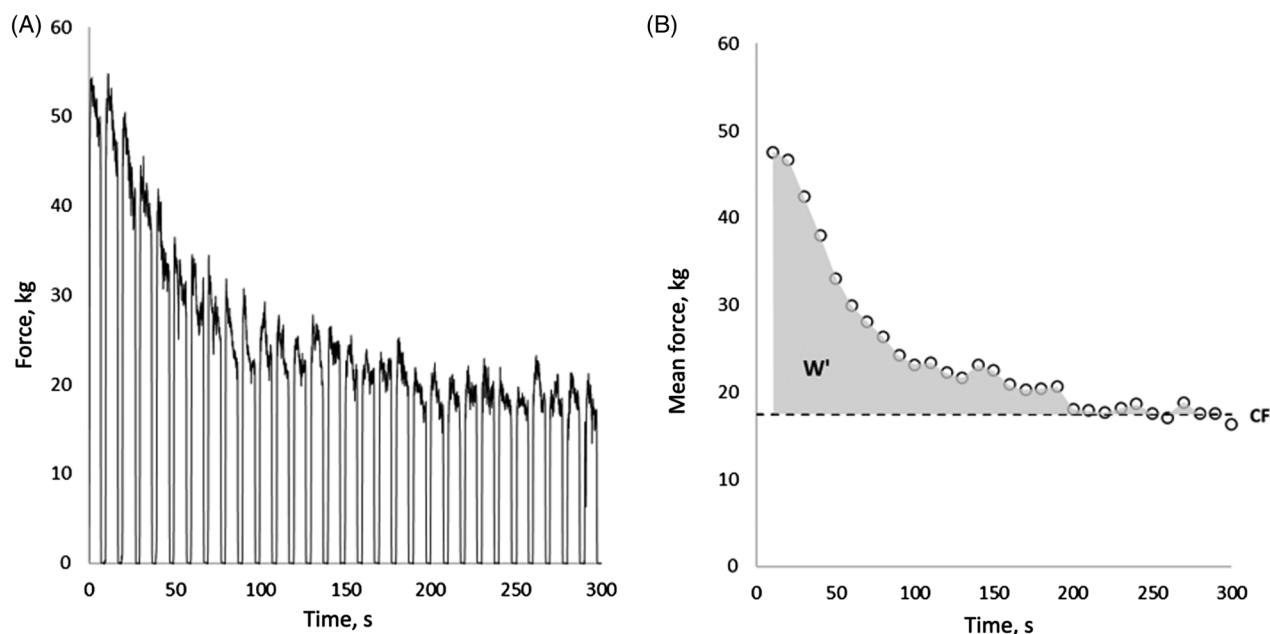


Figure 2 — Force output during a maximal effort CF test in a representative subject. Panel A, raw force trace; panel B, mean contraction force, plotted for all contractions, with CF (in kilograms) shown as a dashed line, representing mean end-test force; and W' (in kilogram second) shown as a gray shaded area, representing the sum of the impulse above end-test force. CF indicates critical force.

the agreement between 4- and 5-minute end-test force was good (intraclass correlation coefficient [ICC] = .650), with only small positive bias (+1.6 kg [limits of agreement of -1.9 to 5.2 kg]). The authors speculate that the positive bias was due to an element of involuntary pacing,²⁰ as the subjects knew the remaining minutes of the test. Given these considerations, the decision was made to use data from the last 60 seconds of each respective test.

The reliability of a similar handgrip ff-CF protocol has been demonstrated by Kellawan and Tschakovsky¹³ with excellent test-retest reliability ($n = 10$) with small within-subject test-retest variation (coefficient of variation 6.8%), small change in the test-retest group mean (typical error 15.3 N, 5.5%), and a high test-retest correlation (Pearson product correlation coefficient [r] = .91, ICC = .94, $P = .01$). The ff-CF pilot data from the authors' lab ($n = 7$) also demonstrated excellent test-retest reliability with small bias (CF, mean difference = -0.03 kg, limits of agreement = -2.45 to 2.4 kg; W' , mean difference = -139 kg·s, limits of agreement = -976 to 698 kg·s) and meet the ICC criterion value (.75) for excellent reliability²¹ (CF ICC = .96; W' ICC = .87).

Statistical Analysis

Normal distributions were ascertained, and homogeneity of variance was confirmed after visual assessment of the frequency histogram and Shapiro-Wilk's test, respectively. All values are reported as mean (SD). All analysis was conducted using the SPSS statistical software package (version 25.0; SPSS Inc, Chicago, IL). To examine the extent to which the physiological responses (CF, CF as a percentage of body mass, W' and W' per kilogram body mass) predicted self-reported sport and boulder-climbing performance, a series of linear regression analyses were performed. Physiological responses were entered into separate models as independent variables both with and without adjustment for the covariate of sex (female coded 1, male coded 0), a combined CF as a percentage body mass and W' per kilogram body mass model was also calculated.

Results

ff-CF Test Characteristics

All subjects were able to complete the CF test in full. In almost all cases, the force decayed to a plateau, a typical example is shown in Figure 2. A small number did not fit this typical pattern, with only a small decline in force across the test and/or considerable end-test variability, these have been excluded ($n = 8$); of note, these subjects were all lower ability with red-point grades of <15 on the IRCRA scale (f6c) and were observed during both the longer 5-minute assessments ($n = 4$) and shorter 4-minute assessments ($n = 4$). Subjects reached the onset of a plateau (defined as within 10% of end-test force) in 156.9 (47.1) seconds.

Association Between ff-CF, W' , and Self-Reported Climbing Ability

Linear regression analysis (Table 2 and Figure 3) revealed that the association between physiological variables and self-reported ability were greatest when expressed as a percentage of body mass, for brevity only these will be discussed. The CF as a percentage of body mass was positively associated with both sport and boulder performance. After adjustment for sex differences, a 1% increase in

CF was associated ($P < .0005$) with an increase of 0.292 (95% CI, 0.236 to 0.347) sport climbing grades (IRCRA) and 0.174 (95% CI, 0.089 to 0.258) boulder-climbing grades (IRCRA). The percentage of variance (R^2) explained were 61% and 26% for sport and boulder disciplines, respectively. The W' per kilogram body mass was not associated with sport-climbing performance but was positively associated with boulder performance. After adjustment for sex differences, a 1 kg·s increase in W' per kilogram body mass was associated ($P < .0005$) with an increase of 0.182 (95% CI, 0.112 to 0.253) in boulder climbing grades (IRCRA). The R^2 explained was 34%. A combined model of CF as a percentage of body mass and W' per kilogram body mass produced significant models for both sport climbing and bouldering; after adjustment for sex differences, the R^2 explained was 66% for sport and 44% for bouldering performance.

Association Between ff-CF, W' , and Peak MIFS

There was a significant association with MIFS peak force and both CF ($R^2 = .301$; $P < .0005$) and W' ($R^2 = .445$; $P < .0005$) illustrated in Figure 4. The strength of the relationship increased for both when sex was entered as a covariate ($R^2 = .585$, $P < .0005$ and $R^2 = .613$, $P < .0005$ for ff-CF and W' , respectively). The strength of the relationship between 40% MVC, a commonly used exercise intensity in climbing training and research, and CF is shown in Figure 4C ($R^2 = .300$; $P < .0005$).

Discussion

The key findings of this study were (1) the ff-CF protocol, comprised of a series of rhythmic isometric maximum voluntary contractions of the finger flexors in a flexed "half-crimp" position, resulted in the same force decay to a plateau that has been seen in isometric critical torque¹⁰ and CF¹³ tests. (2) The association between ff-CF and self-reported climbing ability was excellent, particularly when values were expressed relative to body mass. Combined, ff-CF as a percentage of body mass and W' per kilogram body mass produced significant models for both sport climbing and bouldering; after adjustment for sex differences, the models R^2 were 66% for sport and 44% for bouldering performance. (3) There was a strong positive association between MIFS and both ff-CF and W' . Together, these findings support the use and applicability of an all-out test of ff-CF in climbers for identifying CF and W' .

The decay in force to a plateau across each test was, on the whole, consistent with previous all-out tests across exercise modalities including CP,²² critical speed,²³ critical torque,¹⁰ and CF.¹³ The mean time to a plateau of the subjects in the present study was 157 seconds (~16 contractions). However, of the 129 ff-CF tests conducted, a small number ($n = 8$; 6% of subjects) did not fit the typical pattern, with only a small decline in force and/or considerable end-test variability, and as a consequence, they were excluded from analysis. Notably, these subjects were all lower ability with sport grades of <15 on the IRCRA scale (<f6c) and boulder grades of <19 on the IRCRA scale (<Font 6B+). It would appear that the test is not appropriate for climbers of a lower ability, who are likely to be newer to the sport and who do not currently have the ability to effectively perform a test of this nature. For those who the test is appropriate for, the authors would recommend the use of a 4-minute test when using a 7:3 seconds work to relief ratio, and while it is not possible to predict a suitable test time for other work-relief ratios, it is likely that less frequent and/or shorter contractions will require a longer test to fully deplete W' .

Table 2 Linear Regression Models: Association Between CF (Calculated From End-Test Force), CF as a Percentage of BM (CF %BM), W' (Impulse Measured Above the End-Test Force During the ff-CF Test) and W' per Kilogram BM ($W'/\text{kg BM}$), and Sport and Boulder Climbing Ability

Independent variable	Linear regression models		β	LCI	UCI	P	R^2	Adjusted R^2
Sport red-point performance, IRCRA								
CF	Model 1	Unadjusted	0.397	0.306	0.488	<.0005	.490	.484
	Model 2	Sex (F = 1, M = 0)	0.412	0.309	0.516	<.0005	.493	.480
CF %BM	Model 1	Unadjusted	0.302	0.247	0.357	<.0005	.604	.599
	Model 2	Sex (F = 1, M = 0)	0.292	0.236	0.347	<.0005	.618	.608
W'	Model 1	Unadjusted	0.001	0.000	0.002	.193	.022	.009
	Model 2	Sex (F = 1, M = 0)	0.000	−0.002	0.001	.776	.081	.057
$W'/\text{kg BM}$	Model 1	Unadjusted	0.093	0.000	0.186	.050	.048	.036
	Model 2	Sex (F = 1, M = 0)	0.047	−0.056	0.151	.356	.089	.066
CF %BM and $W'/\text{kg BM}$	Model 1	Unadjusted					.673	.665
		CF %BM	0.308	0.257	0.358	<.0005		
		$W'/\text{kg BM}$	0.112	0.057	0.167	<.0005		
	Model 2	Sex (F = 1, M = 0)					.673	.660
		CF %BM	0.309	0.256	0.361	<.0005		
		$W'/\text{kg BM}$	0.114	0.050	0.178	.001		
Boulder red-point performance, IRCRA								
CF	Model 1	Unadjusted	0.238	0.123	0.353	<.0005	.196	.185
	Model 2	Sex (F = 1, M = 0)	0.200	0.066	0.334	.004	.210	.187
CF %BM	Model 1	Unadjusted	0.194	0.109	0.279	<.0005	.230	.219
	Model 2	Sex (F = 1, M = 0)	0.174	0.089	0.258	<.0005	.284	.263
W'	Model 1	Unadjusted	0.003	0.002	0.004	<.0005	.281	.270
	Model 2	Sex (F = 1, M = 0)	0.002	0.001	0.004	<.0005	.266	.265
$W'/\text{kg BM}$	Model 1	Unadjusted	0.200	0.132	0.269	<.0005	.328	.319
	Model 2	Sex (F = 1, M = 0)	0.182	0.112	0.253	<.0005	.356	.338
CF %BM and $W'/\text{kg BM}$	Model 1	Unadjusted					.454	.439
		CF %BM	0.148	0.074	0.222	<.0005		
		$W'/\text{kg BM}$	0.170	0.107	0.234	<.0005		
	Model 2	Sex (F = 1, M = 0)					.467	.444
		CF %BM	0.140	0.056	0.215	<.0005		
		$W'/\text{kg BM}$	0.160	0.094	0.225	<.0005		

Abbreviations: BM, body mass; CF, critical force; F, female; ff-CF, finger flexor critical force; IRCRA, International Rock Climbing Research Association numeric grading scale; LCI, lower confidence interval; M, male; UCI, upper confidence interval.

The fatigue resistance of the finger flexors is frequently cited as one of the most important factors when describing physical climbing performance along with maximal force production. However, intermittent exhaustive tests conducted at percentages of MVC (typically 40% or 60%) have only been shown to differentiate between climbers of polar abilities (ie, nonclimbers and those of elite ability)^{1,24} but not when there is only a small ability difference,^{24,25} or between disciplines.³ Furthermore, as far as the authors are aware, there is no research looking at the relationship between time to exhaustion during intermittent contractions and self-reported climbing ability. Consequently, not only is the present study the first to demonstrate the feasibility of a single session all-out assessment of ff-CF in climbers, it is also the first to show the extent of the importance of the fatigue resistance of the finger flexors across a wide range of abilities. The combined model, after adjustment for sex differences explained 67% of sport and 47% of bouldering performance (R^2). Differences in the relative contribution of ff-CF and W' were also observed between disciplines, with a greater importance placed on ff-CF in sport climbing and W' in bouldering

performance models. Given the longer, endurance-focused nature of sport climbing the fatigue resistance provided by higher CF is intuitive, likewise a greater energy store component, W' , and relatively lower CF fits the modality of the short, powerful nature of bouldering.

Exhaustive tests using intermittent contractions at percentages of climbers MVC have been used to investigate the exercise capacity of the finger flexors of climbers, with varying success,^{1,3,24,25} as previously discussed. Based on the findings of previous research and the results of the present study, there would be value in using ff-CF or using exercise intensities based on percentages of ff-CF, but not MVC, as the dependent variable in future research for 2 reasons. First, it is well-established that percentages of MVC are not related to metabolic exercise intensity domains.^{26,27} Second, unlike in untrained subjects,¹³ there was a relationship between CF as a percentage of body mass and MVC. As a consequence, while this is unlikely to be a causal relationship, prescribing exercises at an intensity of 40% MVC, typical of many studies, may result in some individuals performing a task

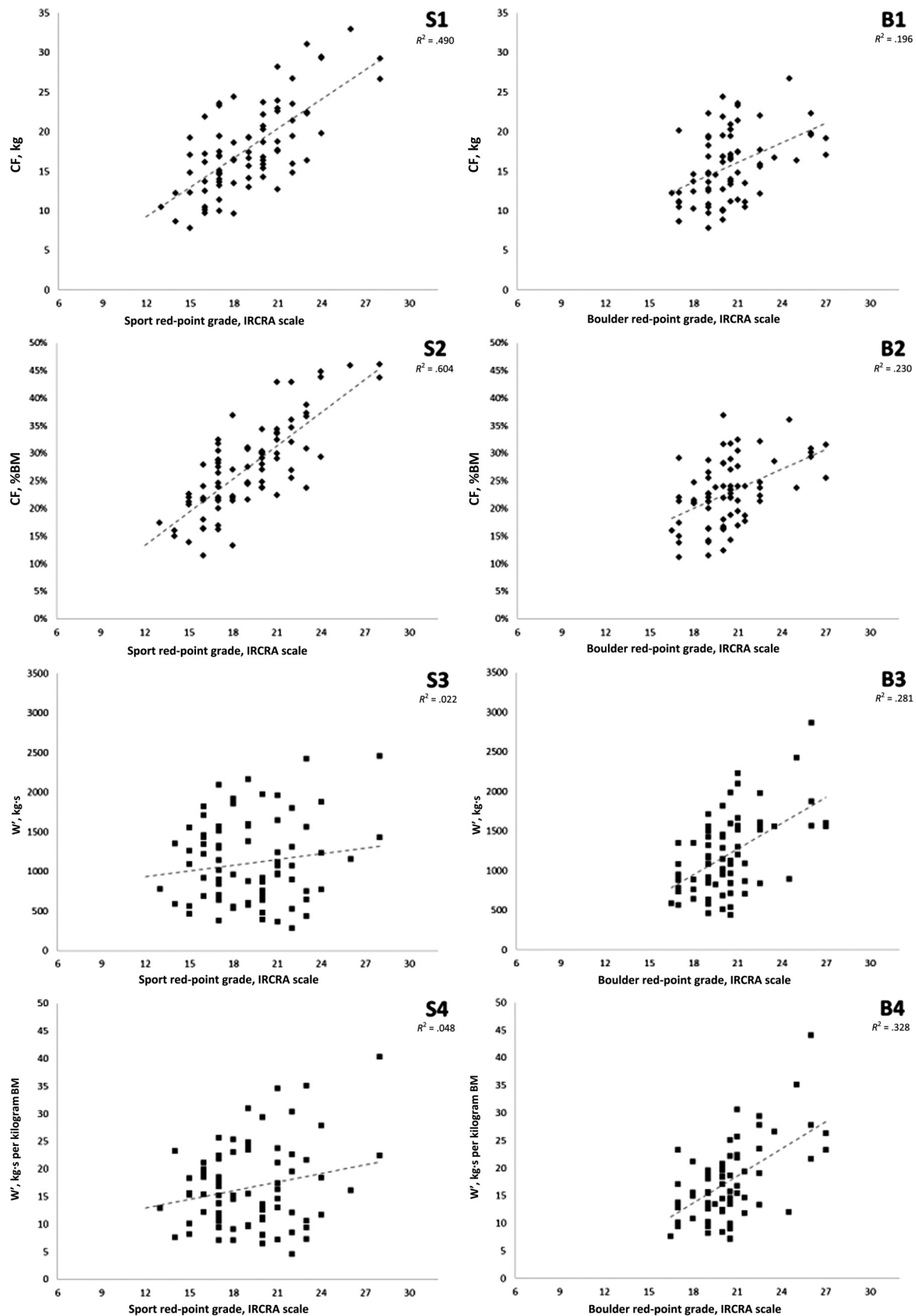


Figure 3 — Linear regression models for CF (calculated from end-test force; panels 1), CF as a percentage of BM (CF %BM; panels 2), W' (impulse measured above the end-test force during the ff-CF test; panels 3), and W' per kilogram BM (W'/kg BM; panels 4) and sport (panel prefix S) and boulder (panel prefix B) climbing ability (IRCRA). BM indicates body mass; CF, critical force; ff-CF, finger flexor critical force; IRCRA, International Rock Climbing Research Association numeric grading scale.

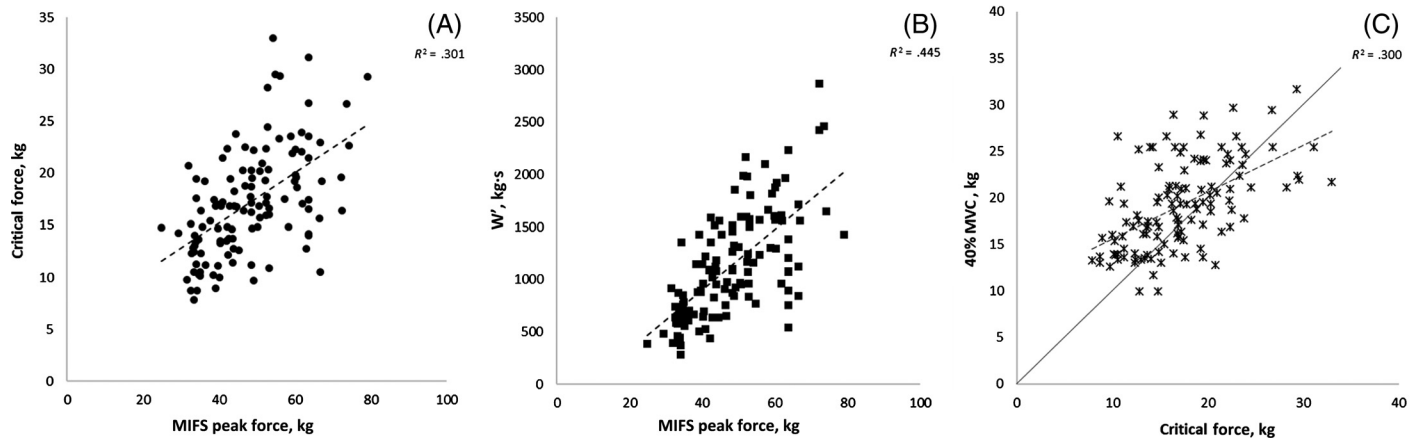


Figure 4 — The association between MIFS peak force and panel A: ff-CF (in kilograms) $R^2 = .301$; $P < .0005$ and panel B: W' (in kilogram second) $R^2 = .445$; $P < .0005$. Panel C: the relationship between the percentages of MVC that ff-CF occurs at and 40% of MVC, solid line is the line of identity where ff-CF = 40% MVC $R^2 = .300$; $P < .0005$. ff-CF indicates finger flexor critical force; MIFS, maximal isometric finger flexor strength.

at intensities that were $<CF$ and others $>CF$ and working in dramatically different exercise intensity domains. The same issue was demonstrated by the authors' previous 2-arm data,⁶ and the present study's single-arm data clearly supports this finding (Figure 4C). Therefore, in the future, ff-CF tests should be used to determine exercise intensities in order to mitigate the potential confounding issue of exercise intensities based on MVCs.

This study has made significant steps in the development of a climbing specific method for the determination of single-arm isometric CF and W' ; however, a number of limitations should be acknowledged. (1) The method described focuses on ff-CF; however, given the climbing-specific testing position employed with the arm extended above the head with the shoulder engaged, it is conceivable that performance is in part also limited by the strength and endurance characteristics of other muscle groups, for example, the biceps and shoulder girdle. There would be value in establishing the relative individual contribution of the finger flexors (eg, using arm fixation) and/or the upper arm and shoulder girdle (eg, using a large hold/edge where the finger flexors are not the limiting factor). (2) Necessarily, the ff-CF test is maximal and performed at a set work–relief ratio; however, this does not wholly represent the varying intensities and contraction lengths seen in the sport. Future research should consider the applicability of a test based on intermittent bouts of exercise.²⁸ Those wishing to employ the described methodology are also reminded that while a 4-minute test is appropriate with 7:3 seconds work–relief ratio, it is likely that less frequent and/or shorter contractions will require a longer test. (3) While the results of the present study present compelling data on the association between self-reported ability and CF and W' , further research is needed to: (1) verify the results presented and their agreement with CF and W' ascertained through multiple constant work rate tests, (2) determine variables that may moderate the strength of the relationship between CF and W' and self-reported ability, and (3) establish the trainability of these characteristics.

Conclusions

The results of the present study demonstrate the applicability of a climbing-specific method for the determination of single-arm isometric CF and W' , highlighting the importance of the fatigue

resistance of the finger flexors in rock-climbing performance. For the majority who were able to reach a stable end-test force, ff-CF as a percentage body mass and W' per kilogram body mass produced significant models for both sport climbing and bouldering. Due to the clear relevance of ff-CF and the inconclusive results of exhaustive tests conducted at percentages of MVC, the authors would recommend that exercise intensities be determined relative to CF. Further research is necessary to determine if ff-CF and W' are trainable characteristics in climbers and the efficacy of interventions based on exercise intensities determined relative to ff-CF end-test force.

Practical Applications

Where the equipment is available, the maximal isometric ff-CF protocol described provides researchers and coaches with a method of assessment that is highly relevant and quick to administer. The authors have previously set out a method of determining CF using equipment that is found in most climbing gyms and where a dynamometer is not available this may be used.⁶ However, the present all-out ff-CF method offers several advantages, including being quicker, possibly more reliable due to the need for only 1 exhaustive test and the ability to also assess contraction duration, impulse, peak, and average force. Due to its demanding nature, the authors recommend ff-CF as an advanced assessment to be completed only with climbers of a higher ability (at least sport f7a and boulder F6C). Finally, given its relevance to both sport climbing and bouldering ability, the authors expect it to become a common test used by coaches for understanding exercise tolerance and for determining optimal training prescription.

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References

- Philippe M, Wegst D, Müller T, Raschner C, Burtscher M. Climbing-specific finger flexor performance and forearm muscle oxygenation in elite male and female sport climbers. *Eur J Appl Physiol*. 2012; 112(8):2839–2847. PubMed ID: [22131087](#) doi:[10.1007/s00421-011-2260-1](#)
- Fryer S, Giles D, Palomino IG, de la O Puerta A, Romero VE. Hemodynamic and cardiorespiratory predictors of sport rock climbing performance. *J Strength Cond Res*. 2017;32(12):3534–3541. doi:[10.1519/JSC.0000000000001860](#)
- Fryer S, Stone KJ, Sveen J, et al. Differences in forearm strength, endurance, and hemodynamic kinetics between male boulderers and lead rock climbers. *Eur J Sport Sci*. 2017;17(9):1177–1183. PubMed ID: [28753391](#) doi:[10.1080/17461391.2017.1353135](#)
- Giles D, Romero VE, Garrido I, et al. Differences in oxygenation kinetics between the dominant and nondominant flexor digitorum profundus in rock climbers. *Int J Sports Physiol Perform*. 2017;12(1): 137–139. PubMed ID: [27145534](#) doi:[10.1123/ijsp.2015-0651](#)
- Baláš J, Mrskoč J, Panáčková M, Draper N. Sport-specific finger flexor strength assessment using electronic scales in sport climbers. *Sports Tech*. 2014;7(3–4):151–158. doi:[10.1080/19346182.2015.1012082](#)
- Giles D, Chidley J, Taylor N, et al. The determination of finger flexor critical force in rock climbers. *Int J Sports Physiol Perform*. 2019;14(7):972–979. PubMed ID: [30676817](#) doi:[10.1123/ijsp.2018-0809](#)
- Poole DC, Ward SA, Gardner GW, Whipp BJ. Metabolic and respiratory profile of the upper limit for prolonged exercise in man. *Ergonomics*. 1988;31(9):1265–1279. PubMed ID: [3191904](#) doi:[10.1080/00140138808966766](#)
- Jones AM, Wilkerson DP, DiMenna F, Fulford J, Poole DC. Muscle metabolic responses to exercise above and below the “critical power” assessed using 31P-MRS. *Am J Physiol Regul Integr Comp Physiol*. 2008;294(2):R585–R593. PubMed ID: [18056980](#) doi:[10.1152/ajpregu.00731.2007](#)
- Coats EM, Rossiter HB, Day JR, et al. Intensity-dependent tolerance to exercise after attaining Vo₂ max in humans. *J Appl Physiol*. 2003; 95(2):483–490. PubMed ID: [12665540](#) doi:[10.1152/japplphysiol.01142.2002](#)
- Burnley M. Estimation of critical torque using intermittent isometric maximal voluntary contractions of the quadriceps in humans. *J Appl Physiol*. 2009;106(3):975–983. PubMed ID: [19150854](#) doi:[10.1152/japplphysiol.91474.2008](#)
- Dekerle J, Vanhatalo A, Burnley M. Determination of critical power from a single test. *Sci Sports*. 2008;23(5):231–238. doi:[10.1016/j.scispo.2007.06.015](#)
- Vanhatalo A, Doust JH, Burnley M. Determination of critical power using a 3-min all-out cycling test. *Med Sci Sports Exerc*. 2007; 39(3):548–555. PubMed ID: [17473782](#) doi:[10.1249/mss.0b013e31802dd3e6](#)
- Kellawan JM, Tschakovsky ME. The single-bout forearm critical force test: a new method to establish forearm aerobic metabolic exercise intensity and capacity. *PLoS One*. 2014;9(4):e93481. PubMed ID: [24699366](#) doi:[10.1371/journal.pone.0093481](#)
- Watts PB, Jensen RL, Gannon E, et al. Forearm EMG during rock climbing differs from EMG during handgrip dynamometry. *Int J Exerc Sci*. 2008;1(1):2.
- Baláš J, Panáčková M, Kodejška J, Cochrane JD, Martin JA. The role of arm position during finger flexor strength measurement in sport climbers. *Int J Perform Anal Sport*. 2014;14(2):345–354. doi:[10.1080/24748668.2014.11868726](#)
- Draper N, Giles D, Schöffl V, et al. Comparative grading scales, statistical analyses, climber descriptors and ability grouping: International Rock Climbing Research Association Position Statement. *Sports Tech*. 2015;8(3–4):88–94. doi:[10.1080/19346182.2015.1107081](#)
- Draper N, Dickson T, Blackwell G, et al. Self-reported ability assessment in rock climbing. *J Sports Sci*. 2011;29(8):851–858. PubMed ID: [21491325](#) doi:[10.1080/02640414.2011.565362](#)
- Torr O, Randall T, Knowles R, Giles D, Atkins S. Reliability and validity of a method for the assessment of sport rock climbers’ isometric finger strength [published online ahead of print March 4, 2020]. *J Strength Cond Res*. doi:[10.1519/jsc.0000000000003548](#)
- Baláš J, Michailov M, Giles D, et al. Active recovery of the finger flexors enhances intermittent handgrip performance in rock climbers. *Eur J Sport Sci*. 2016;16(7):764–772. PubMed ID: [27491378](#) doi:[10.1080/17461391.2015.1119198](#)
- Halperin I, Aboodarda S, Basset F, Byrne J, Behm D. Pacing strategies during repeated maximal voluntary contractions. *Eur J Appl Physiol*. 2014;114(7):1413–1420. PubMed ID: [24658878](#) doi:[10.1007/s00421-014-2872-3](#)
- Fleiss J. *Reliability of measurement. The design and analysis of clinical experiments*. New York, NY: John Wiley & Sons; 1986.
- Cheng CF, Yang YS, Lin HM, Lee CL, Wang CY. Determination of critical power in trained rowers using a three-minute all-out rowing test. *Eur J Appl Physiol*. 2012;112(4):1251–1260. PubMed ID: [21769731](#) doi:[10.1007/s00421-011-2081-2](#)
- Broxterman R, Ade C, Poole DC, Harms CA, Barstow TJ. A single test for the determination of parameters of the speed–time relationship for running. *Respir Physiol Neurobiol*. 2013;185(2):380–385. PubMed ID: [22981969](#) doi:[10.1016/j.resp.2012.08.024](#)
- Fryer SM, Stoner L, Dickson TG, et al. Oxygen recovery kinetics in the forearm flexors of multiple ability groups of rock climbers. *J Strength Cond Res*. 2015;29(6):1633–1639. PubMed ID: [25536538](#) doi:[10.1519/JSC.0000000000000804](#)
- MacLeod D, Sutherland D, Buntin L, et al. Physiological determinants of climbing-specific finger endurance and sport rock climbing performance. *J Sports Sci*. 2007;25(12):1433–1443. PubMed ID: [17786696](#) doi:[10.1080/02640410600944550](#)
- Saugen E, Vollestad N, Gibson H, Martin P, Edwards R. Dissociation between metabolic and contractile responses during intermittent isometric exercise in man. *Exp Physiol*. 1997;82(1):213–226. PubMed ID: [9023519](#) doi:[10.1113/expphysiol.1997.sp004010](#)
- Kent-Braun J, Miller R, Weiner M. Phases of metabolism during progressive exercise to fatigue in human skeletal muscle. *J Appl Physiol*. 1993;75(2):573–580. PubMed ID: [8226454](#) doi:[10.1152/jappl.1993.75.2.573](#)
- Jones AM, Vanhatalo A. The ‘critical power’ concept: applications to sports performance with a focus on intermittent high-intensity exercise. *Sports Med*. 2017;47(suppl 1):65–78. doi:[10.1007/s40279-017-0688-0](#)

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